

Charged Particle in a Uniform Electric Field

Charged Particle in an Electric Field · oPhysics

Senior Secondary (Years 11-12)

~30 minutes

Science · Physics

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Simulation: <https://ophysics.com/em6.html>

Curriculum links: electric fields

Learning intentions

- I can apply $F = qE$ and $a = qE/m$ to a charged particle entering a uniform field.
- I can compare the motion of a charged particle in an electric field to projectile motion under gravity.

Getting started

Open the simulation, set the particle's charge, mass and initial velocity, and set the electric field strength. Run the simulation and observe the shape of the path.

Tasks

1. Predict the general shape of the particle's path before running the simulation. What type of motion does this remind you of?
2. Using your chosen charge, mass and field strength, calculate the particle's acceleration using $a = qE/m$.

Working:

3. Reverse the sign of the charge, keeping everything else the same. Predict which direction the deflection will now be, then check.
4. Keeping the charge and mass constant, run the simulation at four different field strengths and record the deflection at a fixed horizontal distance.

Field strength

Deflection

Extension (optional)

Compare and contrast this simulation with projectile motion under gravity. What plays the role of 'g' here, and what would need to change for the two situations to behave identically?

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